温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Course Review & Final Exam Preview

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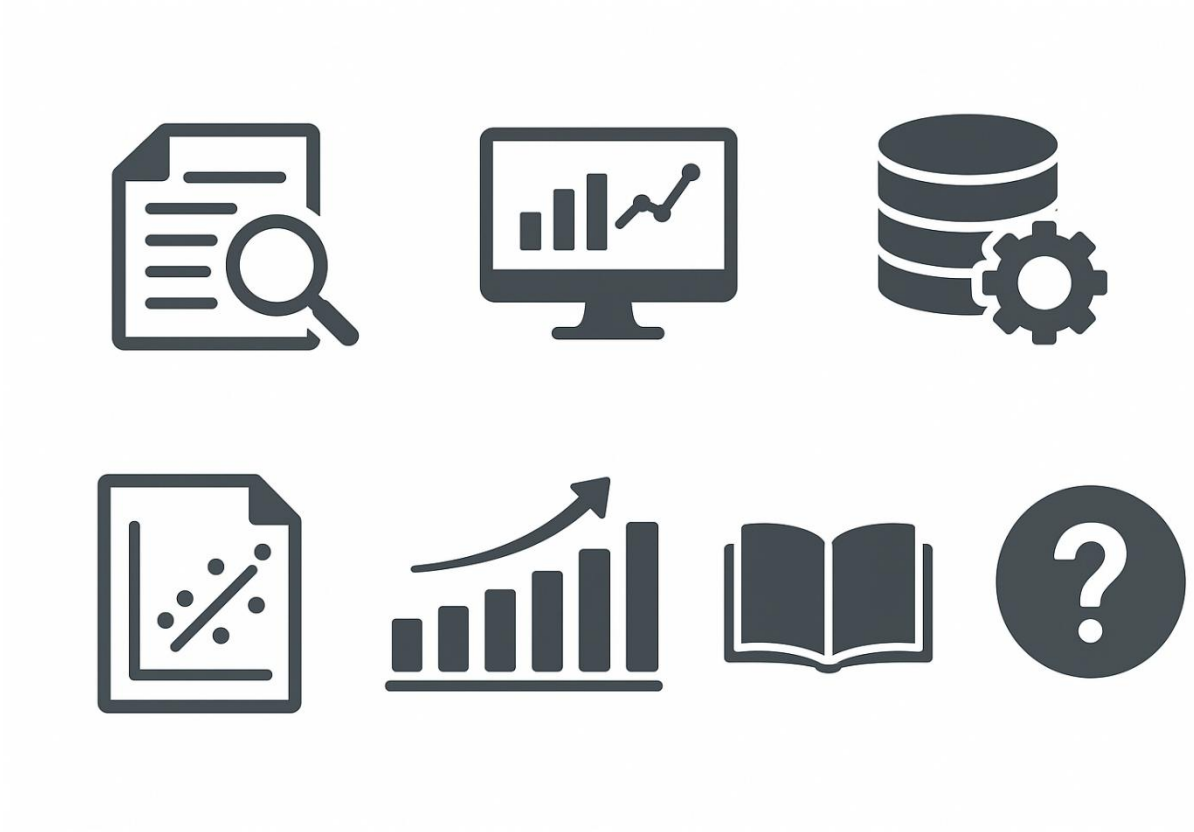
Week 16-1, 16 Dec 2025

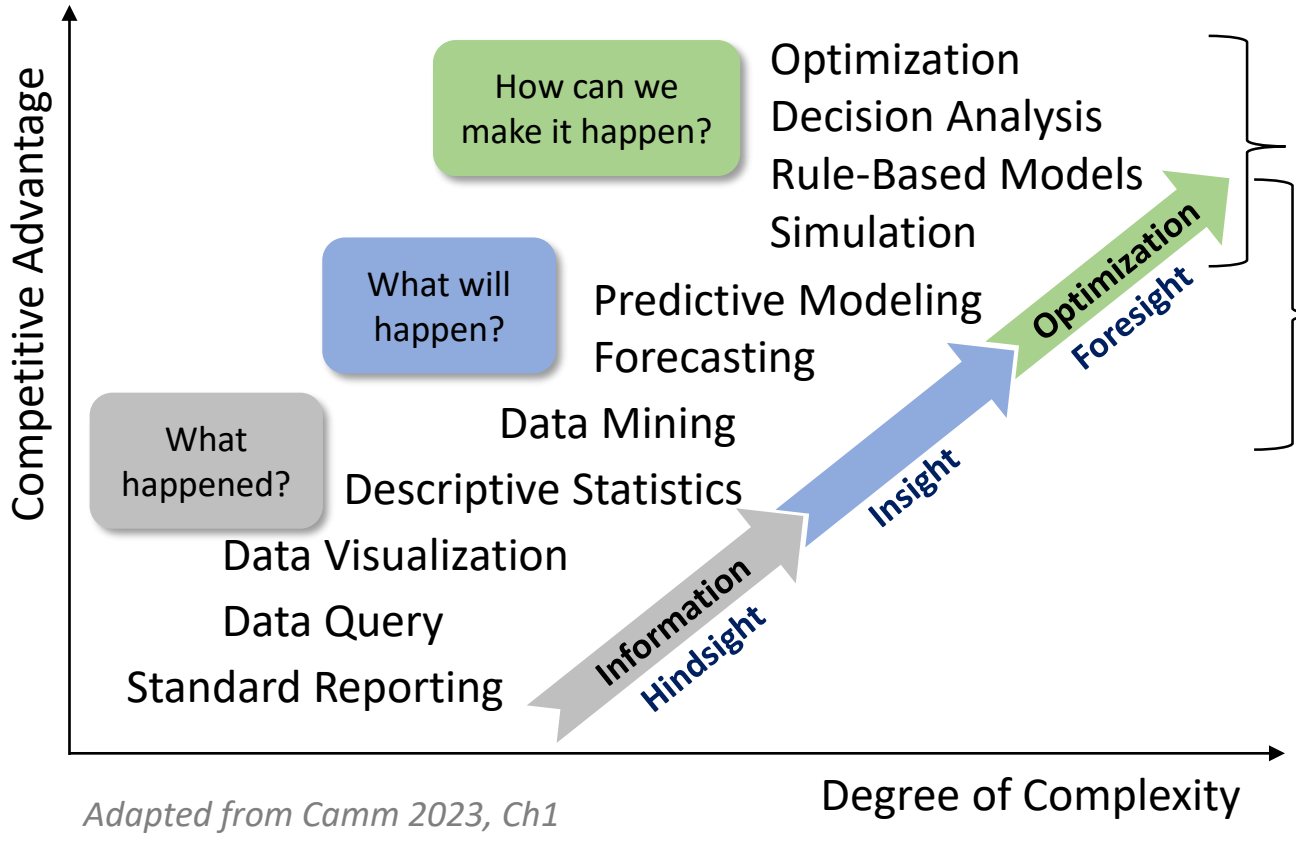
Outline

- Course Review
  - Descriptive analysis
  - Visualization
  - Data wrangling
  - Hypothesis testing
  - Regression analysis
- Final Exam Preview
  - Structure
  - Sample questions



Spectrum of BA

Diagnostic Analytics: why did it happen? Builds on descriptive results by finding correlations, causes, and contributing factors.



Adapted from Camm 2023, Ch1

Prescriptive — Identify the best alternatives to minimize or maximize some objectives

Predictive — Predict future by examining historical data, detecting patterns or relationships in these data, then extrapolating these relationships forward in time

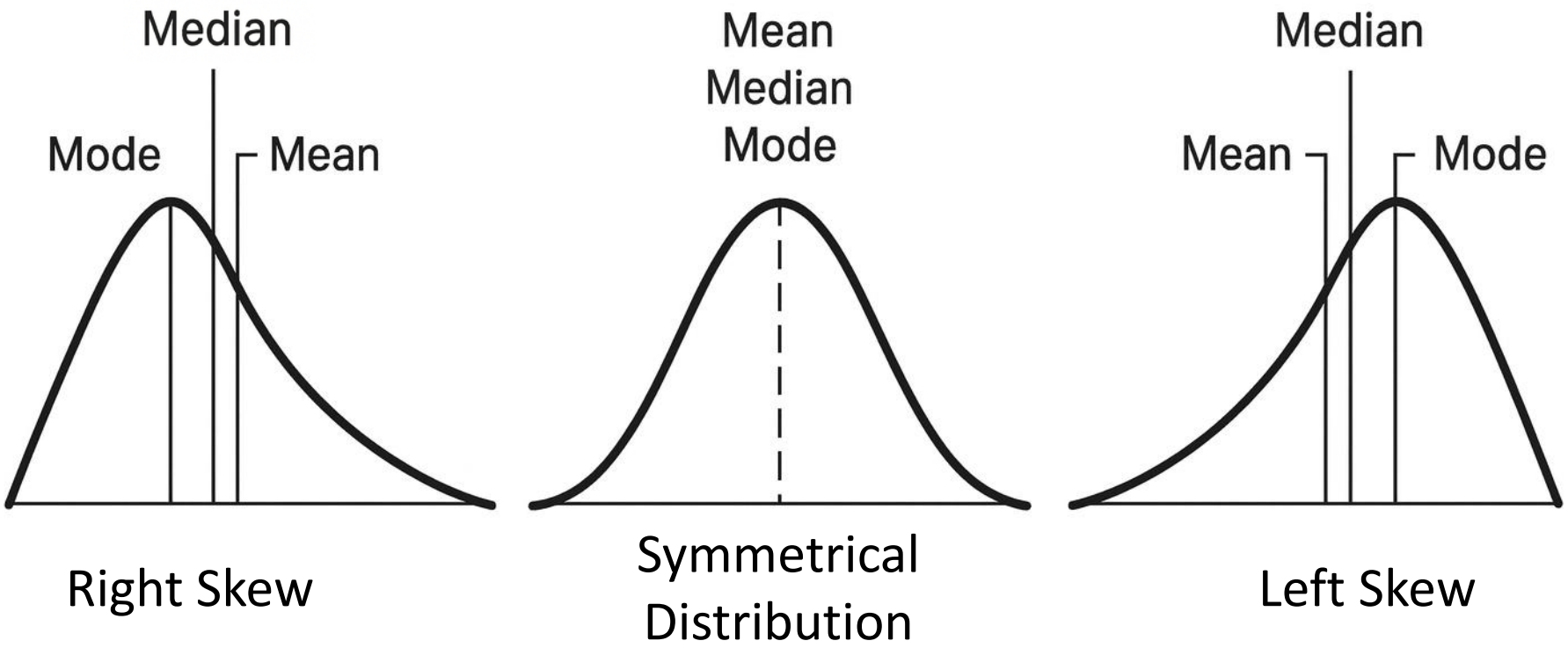
Descriptive — Use of data to understand past & current business performance & make informed decisions

Exploratory Data Analysis (EDA): a starting point, uses visualization and summary statistics to explore patterns, spot anomalies, and generate hypotheses.

Descriptive Analysis

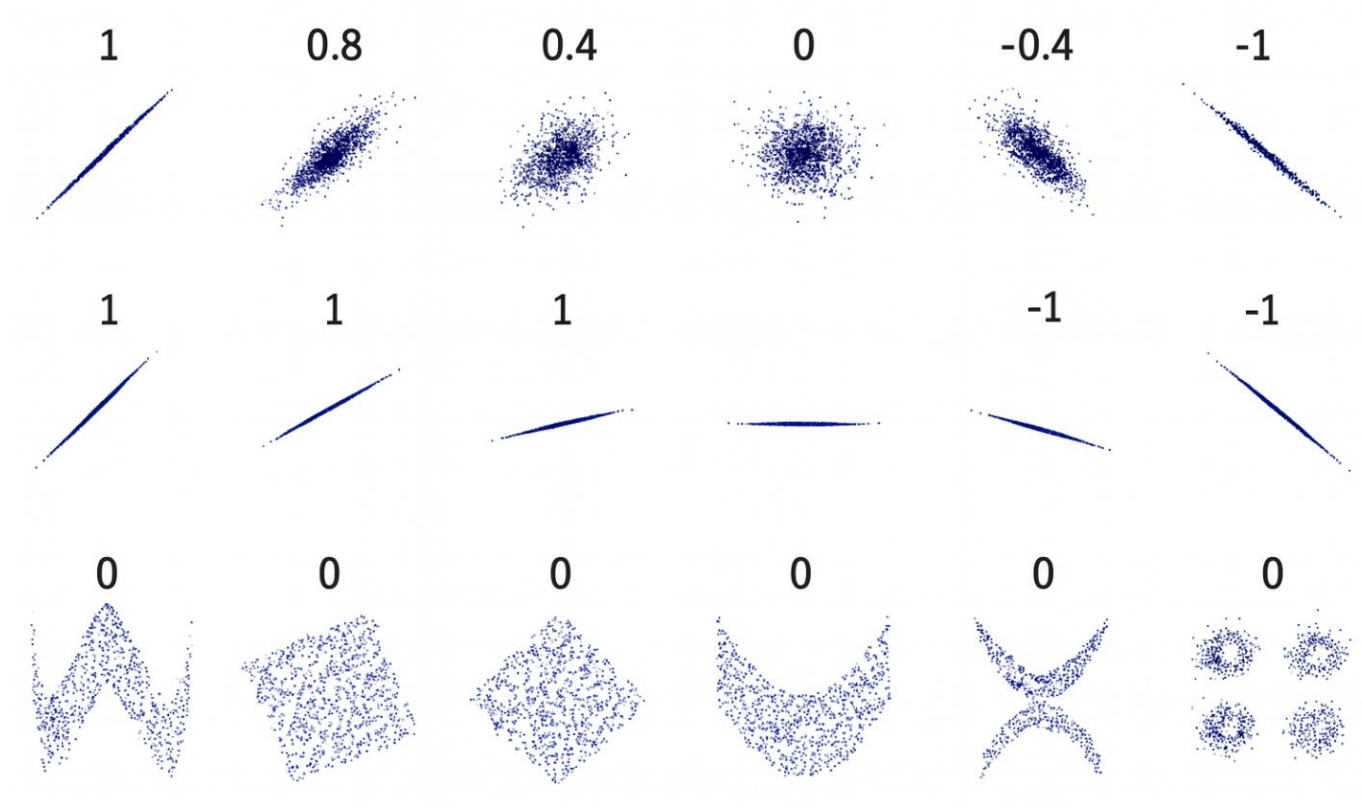
- For single variable
  - Location: mean, median, mode
  - Dispersion: min, max, range, percentile, IQR, variance, std. dev.
  - Shape: skewness, kurtosis
  - Outliner: z-score, IQR
- For two or more variables
  - Covariance: direction of relationship
  - Correlation (matrix): direction and strength of linear relationship

Descriptive Analysis



https://en.wikipedia.org/wiki/Skewness

Descriptive Analysis



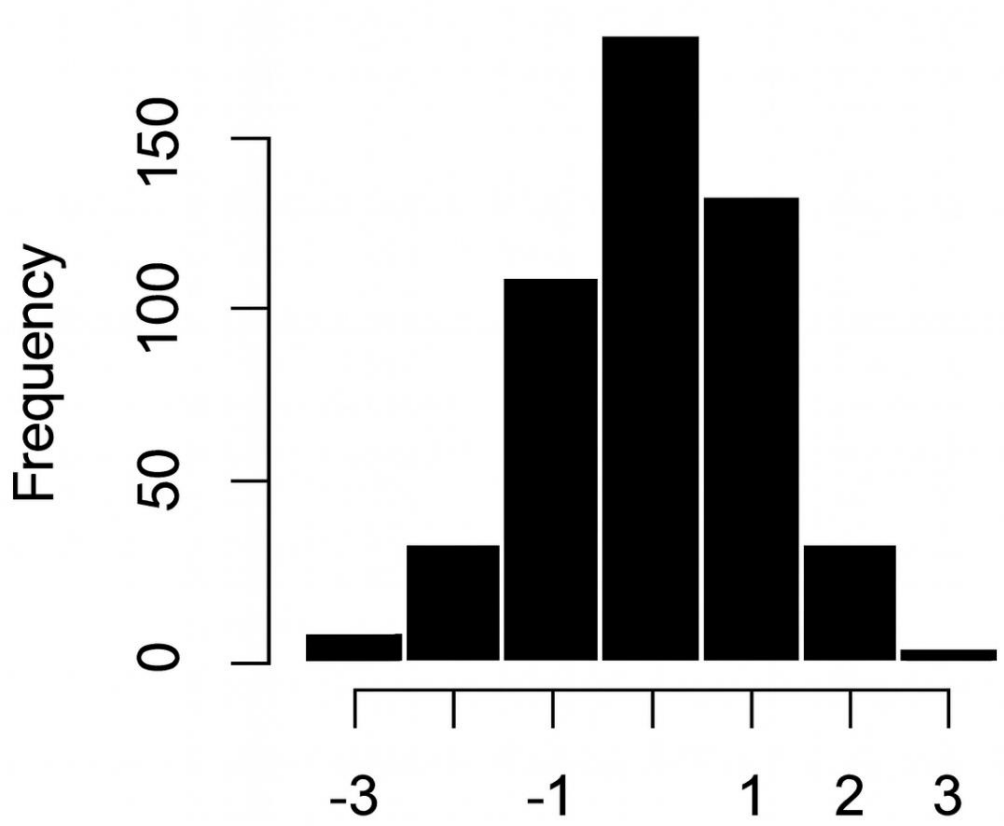
Source: Wikipedia (Correlation)

Visualization

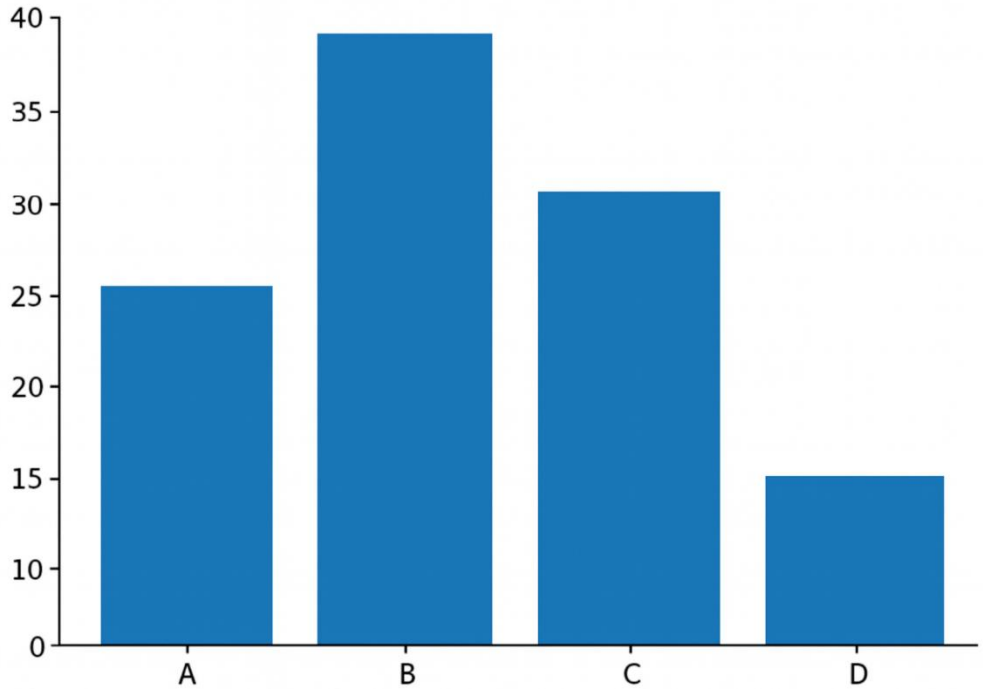
Analysis Purpose	Recommended Chart Types
Compare values across categories	Column chart, bar chart, radar chart
Show **part-to-whole** relationships	Pie chart, donut chart, stacked column chart, tree map
Analyze data **distribution**	Histogram, box plot, scatter plot, density plot, bubble plot, heat map
Display **trend**s over time	Line chart, area chart, candlestick chart
Show **relationships & correlations**	Scatter plot, bubble chart, hear map, correlation matrix
Display **geographic** data	Map, heat map, point map, flow map
Display **hierarchical structure**	Tree map, sunburst chart, Sankey diagram
Show **network & connections**	Network graph, chord diagram, force-directed graph

Visualization

Histogram

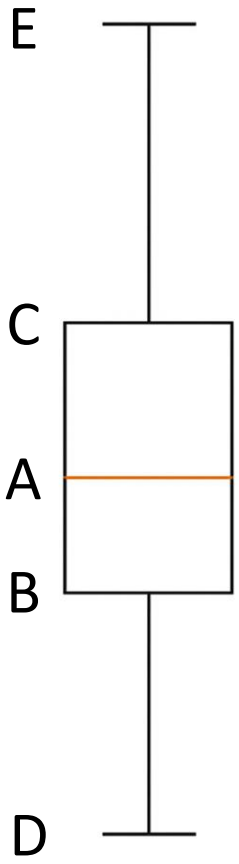


Bar Chart



Visualization

Box Plot



What is A, B, C, D, E?

Data Wrangling

1) Discovering
   - Explore and **understand** raw data – its structure, content, and potential use
2) Structuring
   - **Arrange** raw data into a usable format for analysis
3) Cleaning
   - **Detect** and **correct** errors, inconsistences, missing values, duplicates, etc.
4) Enriching
   - **Add** new information or **transform** existing data to enhance value
5) Validating
   - **Check** accuracy, reliability, and readiness for analysis
6) Publishing
   - **Save** and **share** the clean, well-documented dataset for use

Data Cleaning

- Key tasks
  - Handle **missing values**: detect and decide how to impute
  - Remove **duplicates**: find and drop
  - **Data type** conversions: e.g., convert to datetime
  - **Text cleaning**: trim spaces, unify case, remove symbols, use regex
  - **Categorical** processing: create dummy or ordinal variables
  - **Normalization & scaling**: apply min-max, z-score, or robust scaling (median, IQR)
  - Handle **outliners**: detect and decide to keep or remove (z-score, IQR)
  - Data **transformation**: log, square root, power, binning, etc.
- Tips
  - Identifying and removing any bad data greatly impacts the rest of the wrangling processes.
  - Avoid overcleaning — do not remove useful variation or rare patterns

Hypothesis Testing

Procedure

1) Develop the null and alternative hypotheses (H_0 vs. H_1 or H_a)
2) Specify the level of significance α
3) Collect the sample data and calculate a test statistic
4) Determine the decision rule on which to base a conclusion
5) Critical value, rejection region, p-value
6) Apply the decision rule and draw a conclusion in the context of the application

Hypothesis Testing

One-tailed (Lower-tail)	One-tailed (Upper-tail)	Two-tailed
H_0: $\mu \geq \mu_0$	H_0: $\mu \leq \mu_0$	H_0: $\mu = \mu_0$
H_1: $\mu < \mu_0$	H_1: $\mu > \mu_0$	H_1: $\mu \neq \mu_0$

		Population Condition	
		$\boldsymbol{H_0}$ **True**	$\boldsymbol{H_0}$ **False**
Conclusion you made	**Fail to reject $\boldsymbol{H_0}$**	Correct Inference (true negative) (probability = $1 - \alpha$)	Type II Error (false negative) (probability = β)
	Reject $\boldsymbol{H_0}$	Type I Error (false positive) (probability = α)	Correct Conclusion (true positive) (probability = $1 - \beta$)

Hypothesis Testing

Example: “I will not detect, criticize, or penalize AI use because of false positives.”

1) What is the null hypothesis (H_0) here?
2) What is “false positive” here?
3) What is “false negative” here?
4) Which error is more serious: 误判 (false positive) or 漏判 (false negative)?

Hypothesis Testing

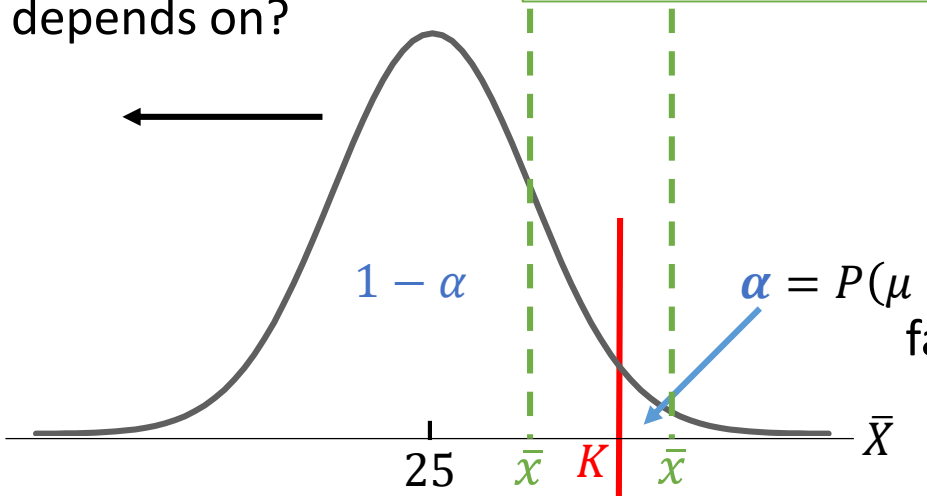
Decision Rule: Reject H_0 if $\bar{x} \geq K$.

<u>Think:</u> **Critical value K** depends on?

What is p-value = $P(\bar{X} \geq \bar{x} | \mu \leq 25)$?
Measure the **fit** between **sample** data and H_0

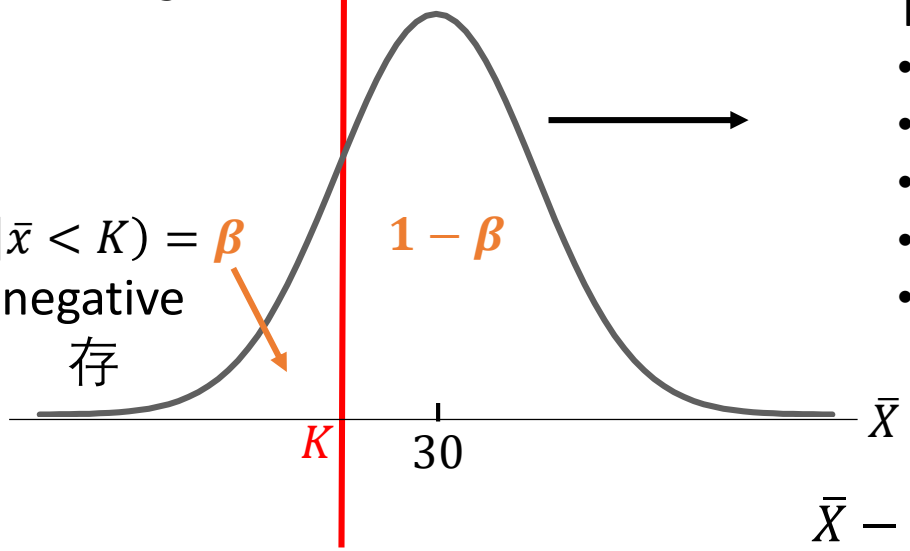
H_0: $\mu \leq 25$

H_1: $\mu > 25$



$$\alpha = P(\mu \leq 25 | \bar{x} \geq K) = \frac{P(\bar{x} \geq K | \mu \leq 25) P(\mu \leq 25)}{P(\bar{x} \geq K)}$$

false positive
真 弃



$$\frac{P(\bar{x} < K | \mu > 25) P(\mu > 25)}{P(\bar{x} < K)} = P(\mu > 25 | \bar{x} < K) = \beta$$

false negative
伪 存

Think:

- Which one is more important: α, β?
- $\alpha \uparrow,\ \beta \downarrow$
- $\alpha + \beta = 1$?
- β depends on population mean.
- How sample size $\boldsymbol{n}$, population std $\boldsymbol{\sigma}$ influence $\boldsymbol{K}$?

$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ $\quad \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$ $\quad t \approx z$ if $n > 30$

Regression Analysis

"True" model: $y = \beta_0 + \beta_1 x + \varepsilon$

Our model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Goal: Choose $\hat{\beta}_0, \hat{\beta}_1$ so as to **minimize the sum of squares of the residuals**

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

OLS Solution:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$$

OLS Estimator

- Random variable
- Sampling distribution
- Unbiasedness, consistency, efficiency

Regression Analysis

A7. Normality (Optional, not Gauss-Markov)

$$\varepsilon_i \sim N(0, \sigma^2)$$

Not needed for BLUE, but need for: exact **t-tests**, exact **confidence intervals**, **F-test**

A1. Linearity in Parameters

The true model is linear in β: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

A2. Random Sampling

(x_i, y_i) are drawn from the population as i.i.d observations.

BLUE

A3. No Perfect Multicollinearity

Regressors must not be perfectly linearly related. In simple regression, x must vary.

A4. Zero Conditional Mean (Exogeneity)

$E[\varepsilon_i | x_i] = 0$, the error term is uncorrelated with x → ensure unbiasedness

A5. Homoscedasticity

$Var(\varepsilon_i | x_i) = \sigma^2$, constant variance of the error term → necessary for OLS estimator to be efficient (minimum variance)

A6. No Autocorrelation

$Cov(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, Error are independent across observations.

Regression Analysis

- When to include a variable
  - Theory / business logic (most important)
  - Statistical evidence: p-value, t-stat, confidence interval
  - Model fit: Adjusted R^2 (penalizes unnecessary IVs)
  - Multicollinearity: VIF
  - Practical significance: effect size, direction, business meaning
- Interpreting regression coefficients
  - Continuous IV (SLR): effect of X on Y
  - Continuous IV (MLR): marginal effect of X on Y holding other IVs constant
  - Quadratic term: nonlinear/diminishing or increasing effect
  - Dummy variable: difference relative to reference group
  - Interaction term: effect of one IV on DV depends on another IV

Final Exam Structure

- **Date:** December 18, 2025 (Thursday)
- **Time:** 20:30 – 21:30 (60 minutes)
- **Venue:** CBPM C226
- **Exam structure** (25 pts)
  - 10 single-choice questions (1 point each)
  - 5 multiple-choice questions (1 point each)
    - Full credit is awarded *only if all correct options are selected*; otherwise, no credit is given.
  - 10 short-answer questions (1 point each)

Final Exam Structure

- Instructions
  - This is a **closed-book exam**, except for <u>one hard copy A4 cheat sheet</u> (both side, printed or written). No electronic devices capable of storing or retrieving text (including electronic dictionaries) are permitted.
  - Do not turn over the examination paper or begin writing until instructed.
  - Record all answers clearly on the **provided answer sheet**.

Single-Choice

A company wants to evaluate the effectiveness of a discount campaign. The dataset `df` contains: *`CustomerID`*, *`UsedDiscount`* (1 = yes, 0 = no), *`Revenue`*. An analyst runs the following code:

```
avg_rev = df[df['UsedDiscount'] == 1]['Revenue'].mean() –
          df[df['UsedDiscount'] == 0]['Revenue'].mean()
print(avg_rev)
```

The analyst concludes: *"Offering discounts increases customer revenue by avg_rev dollars."*

What is the main limitation of this conclusion?

A. The code should calculate median revenue instead of mean revenue
B. The code ignores potential differences between customers who did and did not receive discounts
C. The subtraction order is incorrect and reverses the effect
D. The code must group by *`CustomerID`* before computing revenue

Multiple-Choice

Which of the following statements about multiple linear regression are correct?

A. A regression coefficient represents the effect of a predictor while holding other variables constant

B. Adding more variables always improves the model's causal interpretation

C. Multicollinearity can make coefficient estimates unstable

D. A statistically significant coefficient proves causality

Short-Answer

A company estimates the following regression model to study how **Advertising Spending (A)** and **Price (P)** affect **Weekly Sales**:

$$\widehat{Sales} = 150 + 8A - 0.12A^2 - 5P + 0.24(A \times P)$$

Answer the following questions **<u>briefly</u>**:

a) What does the negative coefficient on A^2 imply about the relationship between advertising spending and weekly sales?

b) The interaction term $A \times P$ has a positive coefficient. Explain what this implies about how the effect of advertising on sales depends on price, and how this insight could inform advertising or pricing decisions.